

# **Software Market Release Notes**

## **eviDense UV Photometer**

### **Instrument Firmware and API Software**



**Instrument manufacturer:**

**Hombrechtikon Systems Engineering AG (HSE)**

**Document number:**

**DMR-DOU-21200612-04907-010\_A2**

**Release date:**

**09.07.2026**

## Table of Contents

|     |                                                   |               |
|-----|---------------------------------------------------|---------------|
| 1   | Document version history .....                    | 3             |
| 2   | About This document .....                         | 3             |
| 3   | Instrument Firmware .....                         | 4             |
| 3.1 | Version History .....                             | 4             |
| 3.2 | What's New in Version 0.8.1 .....                 | 4             |
| 3.3 | Update Recommendation & Impact .....              | 4             |
| 4   | API Software Interface C / CLI.....               | 5             |
| 4.1 | Version History .....                             | 5             |
| 4.2 | What's New in Version 0.8.0 .....                 | 5             |
| 4.3 | Update Recommendation & Impact .....              | 5             |
| 5   | API Software Interface C# .....                   | 6             |
| 5.1 | Version History .....                             | 6             |
| 5.2 | What's New in Version 0.10.1.....                 | 6             |
| 5.3 | Update Recommendation & Impact .....              | <del>76</del> |
| 6   | Compatibility .....                               | <del>87</del> |
| 6.1 | Instrument Firmware / Instrument Hardware .....   | <del>87</del> |
| 6.2 | Instrument Firmware / API Software Interface..... | <del>87</del> |
| 7   | Instructions for Instrument Firmware update.....  | <del>98</del> |
| 7.1 | Required tools.....                               | <del>98</del> |
| 7.2 | Instrument Firmware update procedure .....        | <del>98</del> |

## 1 Document version history

| Document version | Release date | Change                                                                                                                                                                                                                                                              |
|------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 010              | 13.07.2026   | <ul style="list-style-type: none"><li>• Update API Software Interface C#</li><li>• Update Software Interface C</li><li>• Release API Software Interface Python</li></ul>                                                                                            |
| 009              | 13.04.2026   | <ul style="list-style-type: none"><li>• Instrument firmware update 0.8.2</li></ul>                                                                                                                                                                                  |
| 008              | 02.04.2026   | <ul style="list-style-type: none"><li>• Update API Software Interface C#</li></ul>                                                                                                                                                                                  |
| 007              | 30.03.2026   | <ul style="list-style-type: none"><li>• Instrument firmware update 0.8.0</li></ul>                                                                                                                                                                                  |
| 006              | 26.01.2026   | <ul style="list-style-type: none"><li>• API Software Interface C / CLI Version 0.8.0</li><li>• API Software Interface C# 0.10.0</li><li>• Addition of Chapter "Compatibility"</li><li>• Addition of Chapter "Instructions for instrument firmware update"</li></ul> |
| 005              | 02.10.2026   | <ul style="list-style-type: none"><li>• First customer facing version</li></ul>                                                                                                                                                                                     |

## 2 About This document

The eviDense UV Photometer is a compact instrument designed for integration into an automatic liquid handling system.

This document contains the revision history and changes for the eviDense UV Photometer software packages and is intended to give a detailed overview for liquid handler software developers as well as for liquid handler sales and service representatives.

- Instrument firmware: The instrument firmware runs on eviDense UV Photometer instruments and is responsible for the actual optical measurement.
- API Software interface: The API Software interface can be integrated into the liquid handler software and is responsible for obtaining the absorbance values and the calculation of the nucleic acid concentration and purity ratios. The API is available in the following programming languages:
  - C / Command line interface (CLI)
  - C#
  - Python

The API software can be accessed via github: <https://github.com/hseag/evidence>

Incidents and support requests shall be reported to the HSE eviFamily Service Desk:

<https://hseag.atlassian.net/servicedesk/customer/portals>

### 3 Instrument Firmware

#### 3.1 Version History

| Release version | Release date | Changes / Updates                                                                                                                                 |
|-----------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.8.2           | 13.04.26     | Current settling time increased for low current conditions                                                                                        |
| 0.8.0           | 30.03.26     | <ul style="list-style-type: none"><li>Additional information for optical components stored on instrument for internal analysis purposes</li></ul> |
| 0.6.0           | 02.10.25     | <ul style="list-style-type: none"><li>Instrument self-test: dark measurement removed.</li><li>Firmware logging added</li></ul>                    |
| 0.4.0           | 27.02.25     | Initial release                                                                                                                                   |

#### 3.2 What's New in Version 0.8.2

- Release date: 13.04.2026
- Resolved issues: Current settling time increased for low current conditions: Instrument self-test, performed at instrument power-on, could fail due to insufficient current settling time at low LED currents (e.g. 230nm LED). The settling time has been increased from 80 to 150 ms for low-current conditions to ensure proper stabilization and prevent amplifier saturation. This change enables reliable self-test performance and does not impact measurements.
- Improvements / New features: None
- Known issues: None

#### 3.3 Update Recommendation & Impact

- Update recommendation: Update of instruments in field not required.
- Impact: None

## 4 API Software Interface C / CLI

### 4.1 Version History

| Release version | Release date | Changes / Updates                                                                                                                                                                         |
|-----------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.10.0          | 13.07.26     | <ul style="list-style-type: none"><li>A260/A280 purity ratio correction</li><li>Pathlength optimization from 1.1 to 1.04mm</li><li>Absorbance values added to .json output file</li></ul> |
| 0.8.0           | 26.01.26     | <ul style="list-style-type: none"><li>Addition of Run class, in order to facilitate implementation/ integration on liquid handlers</li></ul>                                              |
| 0.6.0           | 18.06.25     | Initial release for end customer usage <ul style="list-style-type: none"><li>Absorbance calculation algorithm updated</li><li>Path length corrected from 1.0 to 1.1 mm</li></ul>          |
| 0.4.0           | 27.02.25     | Initial release for implementation with liquid handler                                                                                                                                    |

### 4.2 What's New in Version 0.10.0

- Release Date: 26.01.2026
- Resolved Issues: None
- Improvements / New features:
  - A260/A280 purity ratio correction: See explanations in Chapter 5.2.
  - Pathlength optimization from 1.1 to 1.04 mm: See explanations in Chapter 5.2.
  - Expansion of the GetResults function to return absorbance values
- Known issues: None

### 4.3 Update Recommendation & Impact

- Update recommendation: Update not required by reseller (HSE internal functionality only)
- Impact: No impact on reseller integration nor on instrument performance

## 5 API Software Interface C#

### 5.1 Version History

| Release version | Release date | Changes / Updates                                                                                                                                                                                                                                                                                                                   |
|-----------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.12.0          | 13.07.26     | <ul style="list-style-type: none"> <li>A260/A280 purity ratio correction</li> <li>Pathlength optimization from 1.1 to 1.04mm</li> <li>Expansion of GetResults function: absorbance values added to .json file</li> <li>Introduction high-level API</li> <li>Additional documentation in github repository</li> </ul>                |
| 0.10.1          | 02.04.26     | <ul style="list-style-type: none"> <li>Update absorbance function to reflect calculation used in concentration determination and purity ratios</li> </ul>                                                                                                                                                                           |
| 0.10.0          | 26.01.26     | <ul style="list-style-type: none"> <li>Functionality added to report progress of firmware update to eviManager Software</li> <li>Addition of Run class, in order to facilitate implementation/ integration on liquid handlers</li> <li>C# implementation made runnable on Linux for HSE unit tests or other future needs</li> </ul> |
| 0.8.1           | 27.06.25     | Initial release for end customer usage <ul style="list-style-type: none"> <li>Typo fixed</li> </ul>                                                                                                                                                                                                                                 |
| 0.8.0           | 18.06.25     | <ul style="list-style-type: none"> <li>Absorbance calculation algorithm updated</li> <li>Path length corrected from 1.0 to 1.1 mm</li> <li>Comments added to NuGet packages for liquid handler SW developer</li> </ul>                                                                                                              |
| 0.6.0           | 16.06.25     | <ul style="list-style-type: none"> <li>Added low level methods for internal HSE QC application</li> <li>Added "GetAvailableDevices" method to receive list of connected eviDense UV modules</li> </ul>                                                                                                                              |
| 0.4.0           | 27.02.25     | Initial release for implementation with liquid handler                                                                                                                                                                                                                                                                              |

### 5.2 What's New in Version 0.12.0

- Release Date: 13.07.2026
- Resolved Issues: None
- Improvements:
  - A260/A280 purity ratio correction: The eviDense UV C# and Python APIs are updated to use the instrument-specific bandpass filter center wavelength to adjust the reported A260/A280 purity ratio. Previously, variations in the center wavelength of the bandpass filter were not taken into account when reporting the A260/A280 purity ratio. As minor variations in the bandpass filter center wavelength can contribute to differences in reported purity ratio values between instruments, this update introduces a correction based on the center wavelength stored on the instrument. The correction minimizes inter-instrument variation and improves the reproducibility and consistency of reported A260/A280 purity ratio results.
  - Pathlength optimization: The optical pathlength has been changed from 1.1mm to 1.04mm to further improve alignment dsDNA concentration alignment with reference standards

- Expansion of the GetResults function to return absorbance values
- Introduction of a high-level API providing a simple workflow
- Documentation: Addition of integration documentation to the GitHub repository describing basic integration steps

### 5.3 Update Recommendation & Impact

- Update recommendation: **An update to version 0.12.0 by resellers is recommended**
- Impact: The change impacts the purity ratio A260/A280 and dsDNA concentration

## 6 API Software Interface Python

### 6.1 Version History

| Release version | Release date | Changes / Updates             |
|-----------------|--------------|-------------------------------|
| 0.10.0          | 13.07.26     | First public released version |

-

## **7 Compatibility**

The following subchapters describe the compatibility between the firmware versions, API versions and the instrument hardware.

### **7.1 Instrument Firmware / Instrument Hardware**

The current instrument firmware is compatible with all released instrument hardware versions.

### **7.2 Instrument Firmware / API Software Interface**

All API versions for C, C# and Python are compatible with all instrument firmware versions.



## **8 Instructions for Instrument Firmware update**

### **8.1 Required tools**

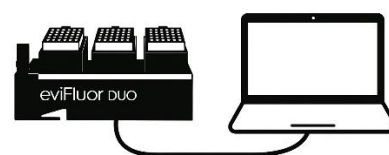
In order to update the instrument firmware, the following is needed:

- eviManager Software
- PC / Laptop
- USB connection cable: USB Type-A to USB Type-C

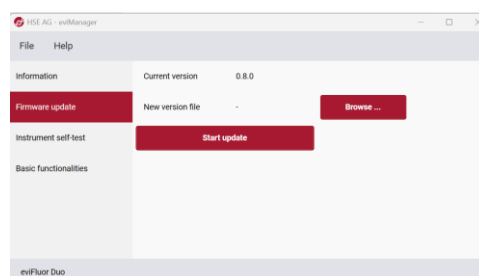
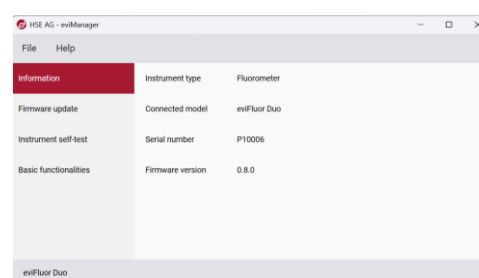
### **8.2 Instrument Firmware update procedure**

(The images and screenshots shown below are taken from the eviFluor Duo instrument and are provided for illustrative purposes only.)

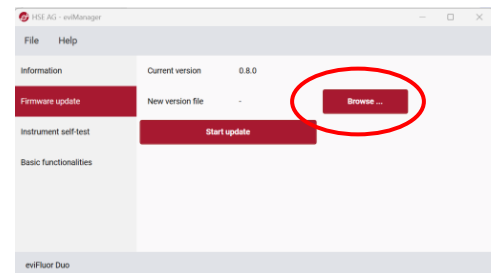
1. Download the latest version of the **eviManager Software** from GitHub:  
<https://github.com/hseag/evimanager>
2. Download the latest firmware version from GitHub and save it to your PC.  
<https://github.com/hseag/evidense/releases/>  
- see respective release, file with extension .srec
3. Connect the eviDense UV instrument to your PC using a USB Type-A to USB Type-C connection cable. (USB Type-C socket on bottom of eviDense UV instrument). Connect the external power supply to the eviDense UV.
4. Open the **eviManager Software** on the PC: The software will automatically detect the connected instrument and display the serial number and current firmware version in the "Information" tab.



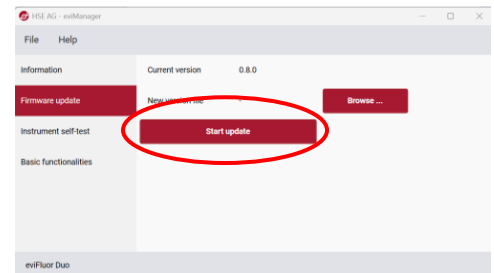
5. Click on the tab "Firmware update"



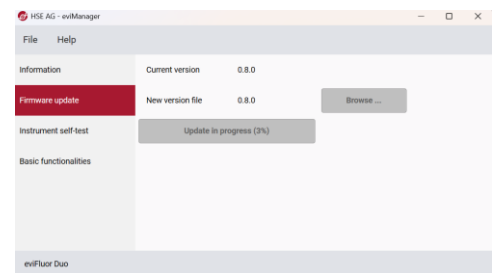
6. Click "Browse" and select the downloaded firmware file



7. Click on "Start update"



8. The update will automatically begin and indicate the progress on the button.



9. Once the update is complete, an on-screen message will confirm successful completion of the update.

